ASDS - 6301 FINAL PROJECT REPORT

A HIERARCHICAL ANALYSIS OF STRUCTURAL BORROWING COSTS IN TEXAS MUNICIPAL DEBT

Kedar Sai Kaustubh Nellimarla (1002331359)

Kishore Vadlamudi (1002332830)

Venkata Satya Sreeya Chandrapati (1002310251)

THE UNIVERSITY OF TEXAS AT ARLINGTON

ABSTRACT

This project investigates the structural drivers of borrowing costs in Texas municipal debt using a hierarchical statistical modeling framework. Municipal debt is a major financing tool for public infrastructure, including schools, utilities, roads, public safety systems, and local government facilities. However, borrowing costs are not determined only by broad market interest rates. They may also vary systematically based on the legal security structure of the bond, the purpose of the issuance, the size of the debt, and the institutional category of the issuing government entity.

Using more than 17,000 Texas local debt issuances, this study defines the **Interest-to-Principal Ratio (IPR)** as the main response variable to measure long-term borrowing burden. The analysis follows a complete data science workflow, including data cleaning, duplicate removal, zero-principal debt filtering, categorical transformation, feature engineering, exploratory data analysis, regression modeling, and hierarchical model evaluation. Exploratory findings show that Revenue bonds generally carry higher IPR values than General Obligation bonds across multiple government categories, indicating a visible "Revenue Premium." The analysis also identifies a non-linear relationship between issuance size and borrowing cost, suggesting that extremely large issuances may face a market saturation effect rather than unlimited economies of scale.

To account for the grouped structure of the data, this project applies a **Hierarchical Linear Model (HLM)** with Government Type as a random intercept. The Intraclass Correlation Coefficient indicates that approximately **6.2% of the variance in borrowing cost is explained by institutional differences across government categories**. Compared with the baseline OLS model, HLM achieves stronger model performance based on AIC and BIC, confirming that municipal debt costs are better modeled using a multilevel structure. The final results show that Revenue bonds carry an estimated **0.06-unit premium in IPR** compared with General Obligation bonds after controlling for issue purpose and principal size. Overall, the study demonstrates that Texas municipal borrowing costs are shaped by both financial characteristics and institutional risk, making hierarchical modeling a more appropriate framework for understanding public debt pricing.

1. INTRODUCTION

Municipal debt plays a central role in financing public infrastructure and essential services. Local governments use debt issuances to fund schools, roads, utilities, public buildings, transportation systems, and other long-term capital projects. In Texas, the municipal debt market includes a wide range of government entities, such as cities, counties, independent school districts, community college districts, water districts, and health-related districts. Although these entities operate within the same state-level debt environment, they may not face the same borrowing costs. Their borrowing burden can vary depending on how the debt is secured, what the debt is used for, and how investors perceive the financial stability of the issuing institution.

A key distinction in municipal finance is the difference between **General Obligation (GO)** bonds and **Revenue (REV)** bonds. General Obligation bonds are typically backed by the taxing authority of the issuer, while Revenue bonds are supported by project-specific or enterprise-specific revenue streams. Because Revenue bonds depend on narrower repayment sources, investors may demand a higher return to compensate for additional repayment risk. This project examines whether that risk premium is visible in Texas municipal debt data and whether it remains significant after controlling other financial and institutional factors.

This analysis also recognizes that municipal debt issuances are not fully independent observations. A loan issued by a school district may share structural similarities with other school district loans, while a water district issuance may follow a different risk profile. This grouped structure violates the independence assumption of a simple linear regression model. Therefore, the project uses a **Hierarchical Linear Model**, also known as a Linear Mixed-Effects Model, to account for institutional clustering by Government Type. This approach allows the model to estimate both overall fixed effects, such as Pledge Type and Issue Purpose, and group-level differences across government categories.

1.1 Problem Statement

Texas local governments issue debt under different legal structures, institutional categories, and financial conditions. However, borrowing costs are often summarized using broad market averages, which may hide important structural differences across issuers. A simple comparison of average interest burden may not reveal whether higher costs are caused by pledge structure, issue purpose, issuance size, or the institutional identity of the borrower.

The main problem addressed in this project is determining whether municipal borrowing costs are primarily market-driven or whether they contain measurable structural patterns. Specifically, the project investigates whether **Revenue bonds carry a higher long-term interest burden than General Obligation bonds** and whether this relationship varies across different government categories. The analysis also examines whether certain government types face systematically higher or lower borrowing costs even after controlling for observable loan characteristics.

1.2 Objective

The primary objective of this project is to quantify the effect of **Pledge Type**, **Issue Purpose**, and **Government Type** on the **Interest-to-Principal Ratio (IPR)** of Texas municipal debt issuances. The IPR is used as the main measure of long-term borrowing burden because it standardizes total interest outstanding relative to total principal outstanding.

The project aims to answer the following research questions:

1. Does a statistically significant **Revenue Premium** exist between Revenue bonds and General Obligation bonds?
2. How much of the variation in borrowing cost is explained by **Government Type**?
3. Does issuance size have a simple linear effect on borrowing cost, or does the relationship show evidence of non-linearity?
4. Does a Hierarchical Linear Model provide a better fit than a standard OLS regression model for this type of grouped municipal debt data?

1.3 Motivation

The motivation for this project is to move beyond surface-level summaries of municipal borrowing costs and identify the structural factors that influence public debt pricing. For local governments, even small differences in borrowing cost can translate into substantial long-term financial consequences. A higher interest burden affects future budgets, taxpayer obligations, project affordability, and fiscal flexibility.

From a statistical perspective, this project is also motivated by the need to model the data using an approach that matches its structure. Since debt issuances are nested within government categories, treating every issuance as completely independent can lead to misleading inference. A hierarchical modeling framework provides a more realistic way to estimate borrowing-cost differences while accounting for institutional clustering.

1.4 Scope

The scope of this project is limited to Texas local municipal debt issuances available in the provided dataset. The analysis focuses on explaining variation in long-term borrowing burden using pledge structure, issue purpose, principal size, and government category. The study includes data preprocessing, feature engineering, exploratory visualization, OLS regression, hierarchical linear modeling, model comparison, and diagnostic evaluation.

This project does not attempt to forecast future interest rates or predict bond market movements. Instead, it focuses on identifying structural borrowing-cost patterns within observed municipal debt records. The final goal is to provide a statistically grounded explanation of how legal security structure and institutional identity influence the cost of public borrowing in Texas.

2. LITERATURE REVIEW

Municipal debt is a major financing tool used by local governments to fund infrastructure projects such as schools, roads, utilities, public buildings, and transportation systems. The cost of this borrowing is influenced by multiple factors, including credit risk, repayment structure, loan size, market conditions, and the institutional profile of the issuing government.

A key idea behind this project is **risk-based bond pricing**. Fama and French explain that bond returns are affected by risk factors related to maturity and default exposure. This supports the expectation that municipal bonds with higher repayment uncertainty should carry higher borrowing costs. In this study, that idea is tested through the difference between **General Obligation (GO)** bonds and **Revenue (REV)** bonds. GO bonds are typically backed by the issuer's taxing authority, while Revenue bonds depend on specific project or enterprise revenues. Because Revenue bonds have a narrower repayment source, investors may require a higher interest burden, creating a possible **Revenue Premium**.

Prior municipal finance research also shows that borrowing costs are affected by issuer characteristics. Capeci found that local fiscal policies and default risk influence municipal borrowing costs, meaning that debt pricing is not purely determined by broad market averages. This supports the use of **Government Type** in this project because cities, counties, school districts, water districts, and community college districts may have different baseline risk profiles.

The literature also supports controlling for issuance size. Larger issuances may benefit from economies of scale and better liquidity, but extremely large issuances may increase concentration risk or investor saturation. This connects with the project's exploratory finding that the relationship between debt size and IPR is not purely linear.

From a statistical perspective, Moulton showed that ignoring grouped data structures can lead to misleading regression inference. Since Texas debt issuances are nested within government categories, a standard OLS model may underestimate uncertainty. Hox's work on multilevel modeling supports the use of a **Hierarchical Linear Model**, where each government category can have its own baseline borrowing-cost level while still estimating the overall effects of pledge type, issue purpose, and principal size.

Overall, the literature supports this project's approach: municipal borrowing costs should be modeled as a combination of financial structure, issuer characteristics, and grouped institutional effects rather than as simple market averages.

3. METHODOLOGY & DATA PREPROCESSING

This section explains how the raw Texas municipal debt dataset was prepared for statistical analysis and hierarchical modeling. The main goal of preprocessing was to make sure that each record represented a valid debt issuance, the financial variables were correctly formatted, and the final dataset was suitable for modeling the **Interest-to-Principal Ratio (IPR)** as the response variable.

3.1 Data Description

The dataset contains Texas local debt issuance records with approximately **17,700 raw observations** and 12 variables. Each row represents a municipal debt issuance, and the dataset includes both institutional and financial information. The major variables include **Government Type**, **Government Name**, **Issuer Name**, **Issuance Name**, **Issue Purpose**, **Closing Date**, **Last Principal Payment Date**, **Fiscal Year**, **Pledge Type**, **Total Principal Outstanding**, **Total Interest Outstanding**, and **Total Debt Service Outstanding**.

The key response variable used in this project is the **Interest-to-Principal Ratio (IPR)**, which measures the long-term interest burden of each issuance relative to its principal amount. The main explanatory variables are **Pledge Type**, **Issue Purpose**, and **Total Principal Outstanding**, while **Government Type** is used as the grouping variable for hierarchical modeling. This structure allows the analysis to capture both issuance-level effects and institutional differences across government categories.

3.2 Data Cleaning & Integrity

The first preprocessing step was a structural integrity audit. Column names were cleaned into a consistent snake_case format to make the dataset easier to process in R. The initial dataset contained **17,717 rows and 12 columns**, with a mixture of character, numeric, and date-related variables.

Next, duplicate records were checked using a composite key made from **Issuance Name**, **Closing Date**, and **Total Principal Outstanding**. This step identified **12 duplicate records**, which were removed to avoid pseudo-replication. Removing duplicates was important because repeated records could artificially inflate the sample size and bias the estimated effects of pledge type or issue purpose.

Date variables such as closing_date and last_principal_payment were converted into proper date format using mdy(). Categorical variables including government_type, pledge_type, and issue_purpose were converted into factors so that the regression and mixed-effects models would treat them as categorical predictors rather than plain text. Financial variables such as total principal and total interest outstanding were converted into numeric format to support ratio construction and regression modeling.

The dataset was also filtered to remove records with zero principal outstanding. These records were treated as invalid "zombie debt" because IPR requires principal in the denominator. Keeping zero-principal records would create undefined or infinite ratio values and distort the analysis.

3.3 Outlier Treatment and Financial Ratio Filtering

After cleaning the core dataset, the primary financial outcome variable was created:

$$IPR = \frac{Total\ Interest\ Outstanding}{Total\ Principal\ Outstanding}$$

This ratio standardizes borrowing cost across issuances of different sizes. A large bond and a small bond may have very different total interest amounts, but IPR allows them to be compared on a common scale.

After calculating IPR, the dataset was filtered to keep only valid and realistic observations. Records with non-positive IPR values were removed because they do not represent meaningful long-term interest burden. Extremely high IPR values above **2.0** were also removed because they represent rare or abnormal cases that could overly influence the regression estimates. This filtering step helped reduce leverage from extreme observations while preserving the main structure of the municipal debt market.

This section should be titled **Outlier Treatment and Financial Ratio Filtering**, not "Outlier and Sensor Repair," because the project does not involve sensor data.

3.4 Feature Engineering

The main engineered feature was the **Interest-to-Principal Ratio (IPR)**, which became the dependent variable for the analysis. This variable captures the relative interest burden of each municipal debt issuance and makes borrowing costs comparable across different issuance sizes.

A second engineered feature was **log principal**:

$$log\_principal = log(Total\ Principal\ Outstanding)$$

This transformation was used because principal amounts in municipal debt data are highly skewed. Some issuances are relatively small, while others are extremely large. Taking the natural logarithm reduces the effect of extreme scale differences and allows the model to capture the relationship between issuance size and borrowing cost more realistically.

The categorical variables were also structured for modeling. **Pledge Type** was used to compare General Obligation, Revenue, and Lease Purchase structures. **Issue Purpose** was included to control for differences in why the debt was issued, such as refunding, transportation, water-related projects, public safety, or general-purpose financing. **Government Type** was used as the Level-2 grouping variable in the Hierarchical Linear Model because issuances from the same type of government may share similar institutional risk characteristics.

Unlike a time-series forecasting project, this analysis does not require lag variables, rolling statistics, Prophet regressors, or holiday indicators. The feature engineering is focused on financial normalization, categorical structure, and hierarchical grouping.

3.5 Modeling Preparation

After preprocessing and feature engineering, the final modeling dataset contained valid municipal debt issuances with complete financial and institutional information. The cleaned dataset was then used for three major analytical steps:

First, exploratory analysis was performed to understand the distribution of IPR, compare borrowing costs across pledge types, and inspect variation across government categories.

Second, a baseline **OLS regression model** was estimated to provide a population-average benchmark. This model helped measure the basic relationship between pledge type, issue purpose, principal size, and IPR.

Third, a **Hierarchical Linear Model** was fitted using Government Type as a random intercept. This model was selected because debt issuances are nested within government categories, and the ICC result showed that institutional grouping explains a meaningful portion of borrowing-cost variation. The final HLM structure allows the analysis to estimate both fixed effects and government-type-level differences.

4. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis was conducted to understand the distribution of borrowing costs, identify structural differences across pledge types and government categories, and evaluate whether the dataset was suitable for regression and hierarchical modeling. The EDA focused on the project's main response variable, the **Interest-to-Principal Ratio (IPR)**, along with key explanatory variables such as **Pledge Type**, **Government Type**, **Issue Purpose**, and **Log Principal**. The purpose of this section is not only to describe patterns visually, but also to justify the modeling decisions used later in the project.

4.1 Distribution of Interest-to-Principal Ratio

The first diagnostic step examined the distribution of the **Interest-to-Principal Ratio (IPR)** using a histogram and a Q-Q plot. The histogram shows that IPR is positively skewed, meaning most debt issuances have relatively moderate interest burdens, while a smaller number of issuances carry much higher long-term interest obligations. This pattern is common in financial ratio data because borrowing costs are usually concentrated around a typical range, but some issuances may have unusually long maturities, higher-risk structures, or unfavorable market timing.

The Q-Q plot confirms that the IPR distribution does not perfectly follow a normal distribution, especially in the upper tail. The deviation at higher quantiles indicates the presence of heavy-tailed behavior. This finding is important because a simple OLS model may be sensitive to non-normality and extreme values. However, after filtering invalid and extreme IPR observations, the dataset remains suitable for regression-based modeling, especially given the large sample size of more than 17,000 issuances.

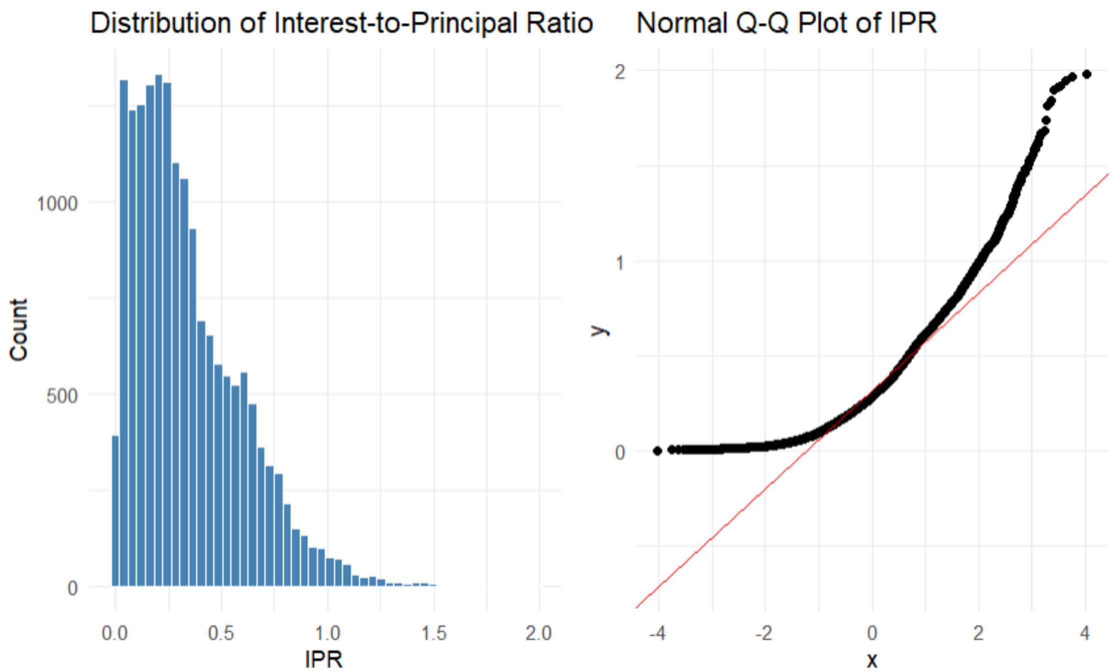


Figure 1:Distribution of Interest-to-Principal Ratio

4.2 Interest Burden by Government Type and Pledge Type

The boxplot comparing IPR across **Government Type** and **Pledge Type** provides the first visual evidence of a structural borrowing-cost difference. Across several government categories, **Revenue (REV)** bonds show higher median IPR values than **General Obligation (GO)** bonds. This supports the project's main hypothesis that Revenue-backed debt carries a higher interest burden because repayment depends on narrower revenue streams rather than broad taxing authority.

The plot also shows that borrowing costs differ across government categories. For example, cities, counties, school districts, community college districts, and water districts do not share the same baseline IPR distribution. This grouped variation suggests that municipal debt issuances are not fully independent observations. Loans issued by the same type of government may share institutional characteristics, legal constraints, or investor perceptions. This visual pattern strongly supports the use of a **Hierarchical Linear Model** with Government Type as a random intercept.

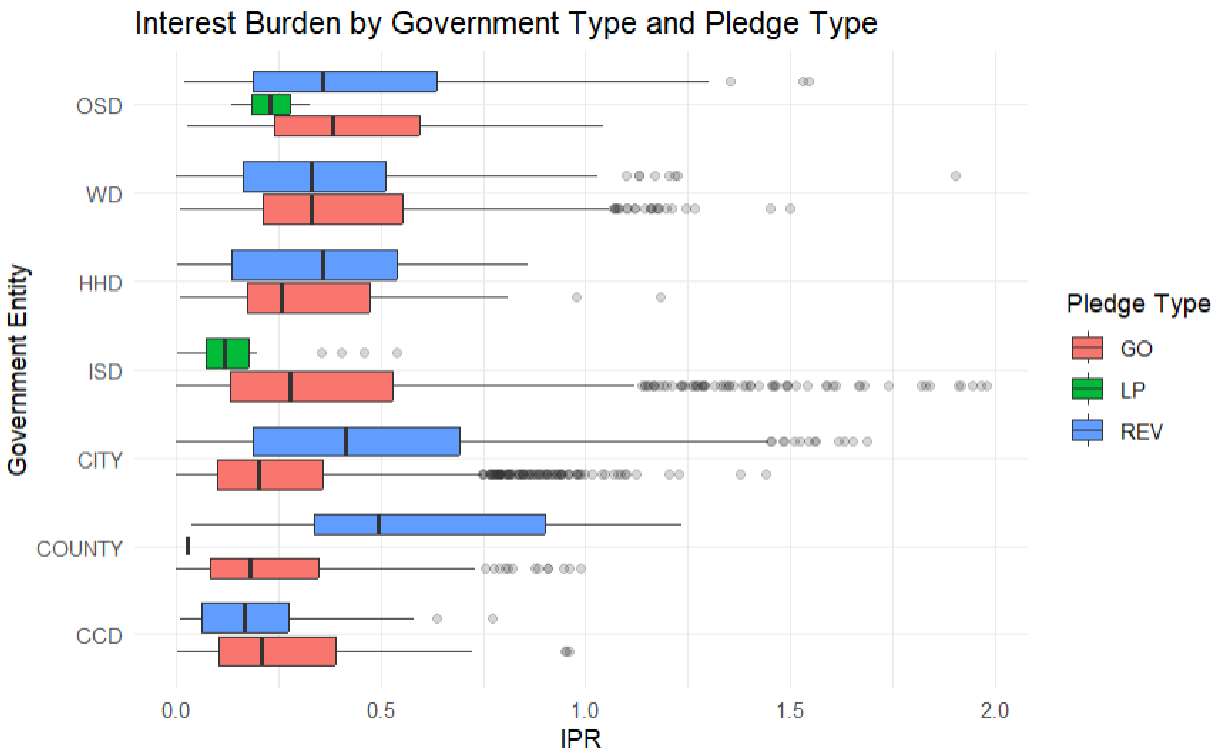


Figure 2: Interest Burden by Government Type and Pledge Type

4.3 Economies of Scale and Market Saturation

The scatter plot of **IPR versus Log Principal**, combined with a LOESS smoothing curve, was used to examine whether larger debt issuances benefit from economies of scale. At lower and moderate issuance sizes, the curve suggests that interest burden may decline as principal size increases. This pattern aligns with the idea that larger issuances can attract more investors, improve liquidity, and spread fixed issuance costs over a larger borrowing base.

However, the relationship is not purely linear. At very high principal levels, the LOESS curve begins to rise again, suggesting a possible **market saturation effect**. This means that extremely large issuances may no longer benefit from scale advantages and may instead require a higher interest burden to attract enough investor demand. This finding is important because it shows that the effect of issuance size is more complex than a simple "larger is cheaper" assumption. It also justifies controlling for principal size in the regression model.

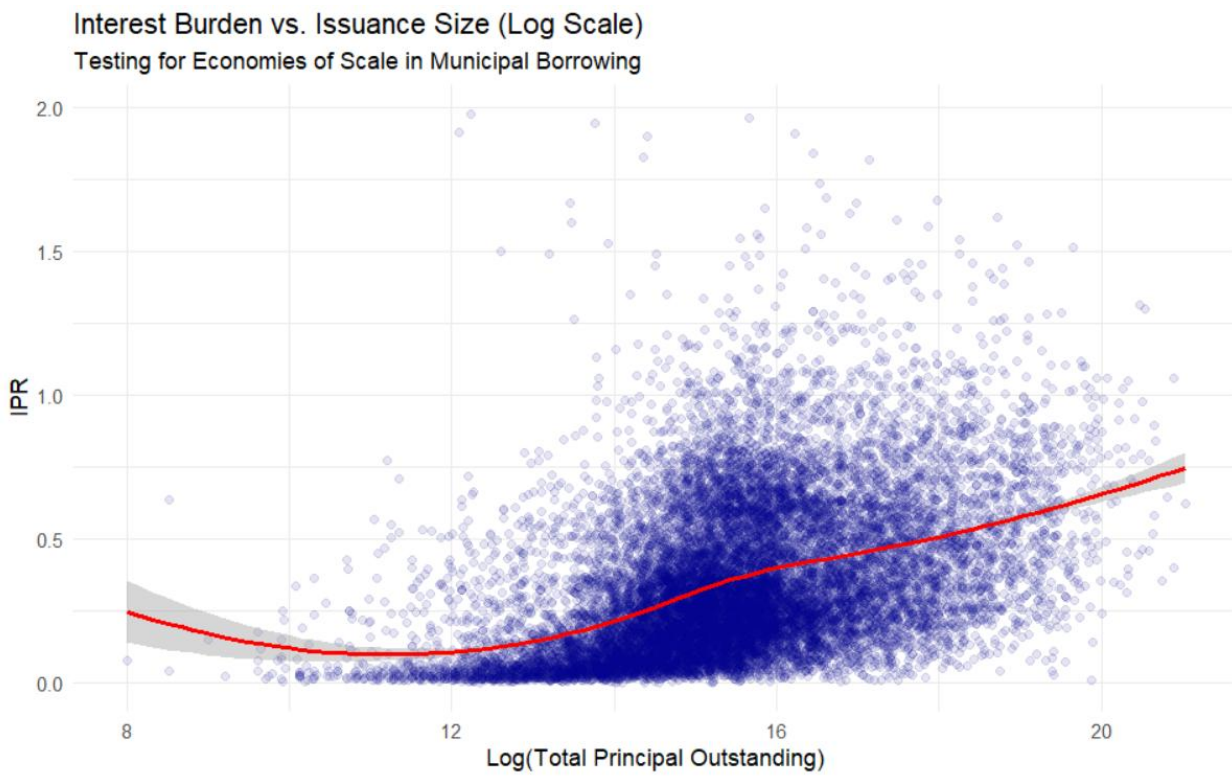


Figure 3: Interest Burdern vs Issuance Size

4.4 Temporal Trends in Borrowing Cost

The time trend plot of mean IPR by **Closing Year** and **Pledge Type** shows how borrowing costs changed over time. The most noticeable pattern is the sharp increase in average IPR after 2021, which reflects the higher-rate borrowing environment after the post-pandemic inflationary period and Federal Reserve rate increases. This confirms that municipal borrowing costs are influenced not only by issuer characteristics but also by broader macroeconomic conditions.

The plot also shows that Revenue bonds generally remain above General Obligation bonds over time. This persistent gap suggests that the Revenue Premium is not limited to one specific year or temporary market condition. Instead, the difference appears to be structural. Even when the overall borrowing-cost environment shifts upward, Revenue bonds continue to carry a higher interest burden than GO bonds.

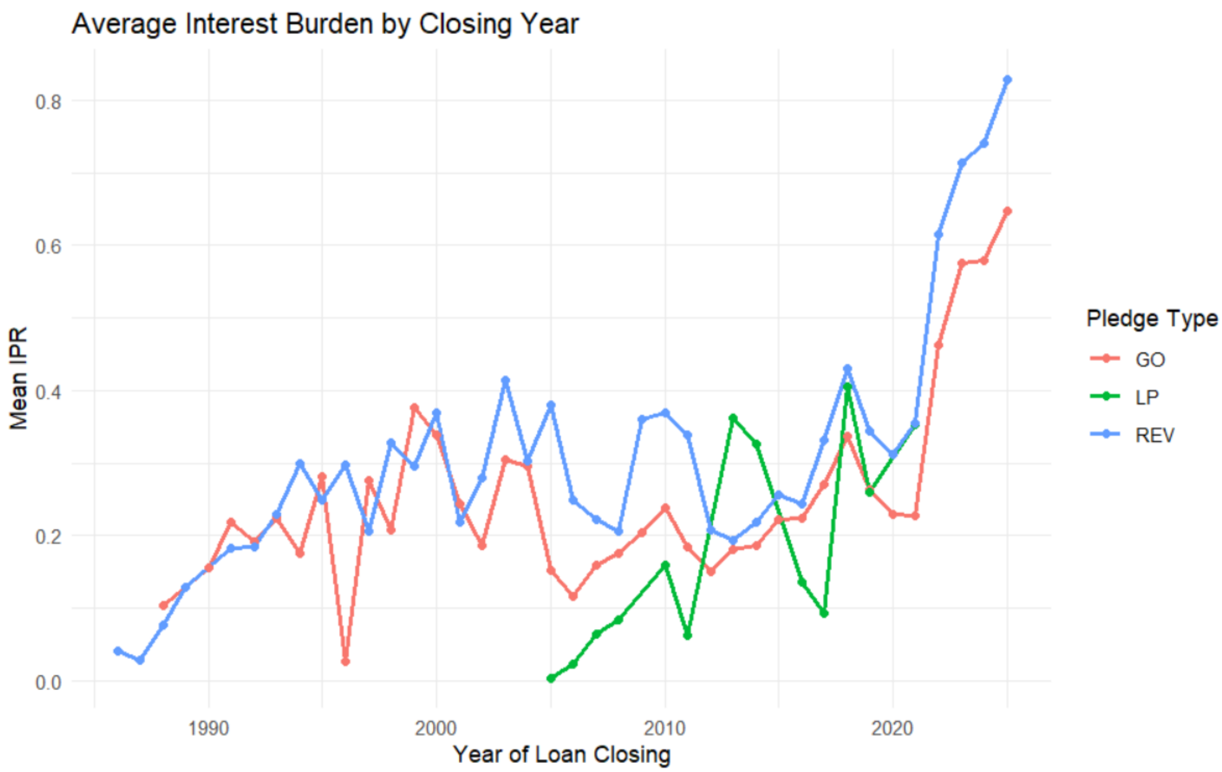


Figure 4: Average Interest Burdern by Closing Year

4.5 Structural Overlap and Collinearity Risk

A heatmap of **Issue Purpose versus Pledge Type** was used to check whether certain purposes were strongly tied to specific pledge structures. This step was important because if a purpose category appears almost exclusively under one pledge type, the model may struggle to separate the effect of pledge structure from the effect of issue purpose.

The heatmap shows that some issue purposes are more commonly associated with specific pledge types. For example, certain infrastructure or utility-related purposes may rely more heavily on Revenue pledges, while other public-purpose categories may appear more often under GO pledges. This creates a potential risk of structural overlaps. However, the later GVIF diagnostic shows that the adjusted GVIF values are close to 1 and below the conservative threshold of 2, indicating that multicollinearity is not severe enough to destabilize the model estimates.

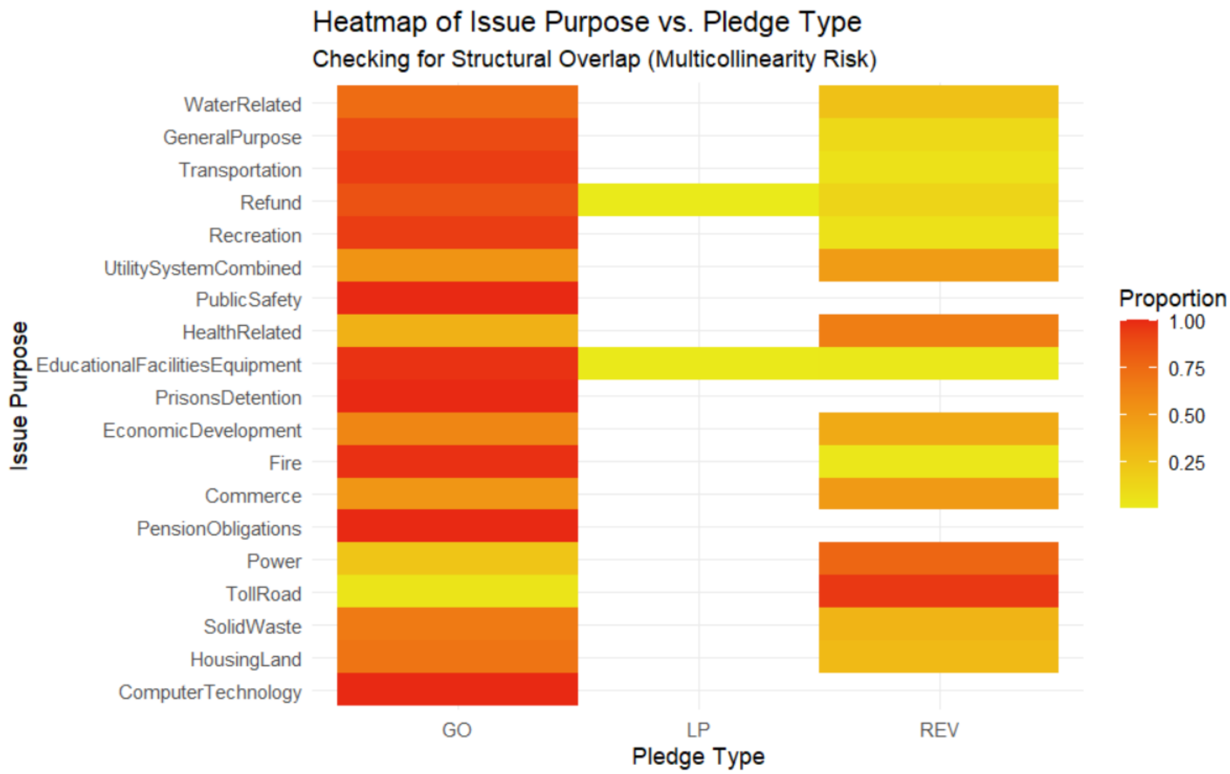


Figure 5: Heatmap of Issue Purpose vs Pledge Type

4.6 Pre-Modeling Diagnostics

Before fitting the final models, three additional diagnostics were used to evaluate whether the dataset was appropriate for hierarchical modeling. First, the group count by **Government Type** showed that the Level-2 grouping variable was sufficiently populated. Although some government categories had more observations than others, even the smaller groups had enough records to support estimation of group-level variation.

Second, the leverage and outlier distribution plot showed that high-IPR observations were not concentrated in only one government category. Instead, the data contained a dense distribution of observations across groups, with some long-tailed behavior. This suggests that the model should account for group-level differences rather than treating the entire dataset as one homogeneous population.

Third, the GVIF multicollinearity diagnostic confirmed that the major predictors were not highly redundant. The adjusted GVIF values for **Pledge Type**, **Issue Purpose**, and **Log Principal** were near 1 and below the threshold of concern. This result is important because it means the model can estimate the Revenue Premium separately from the effects of issue purpose and principal size.

Overall, the EDA shows four major patterns: Revenue bonds carry visibly higher interest burdens than GO bonds, government categories have different baseline borrowing profiles, issuance size has a non-linear relationship with borrowing cost, and the predictor structure is stable enough for regression modeling. These findings directly support the transition from exploratory analysis to a Hierarchical Linear Model.

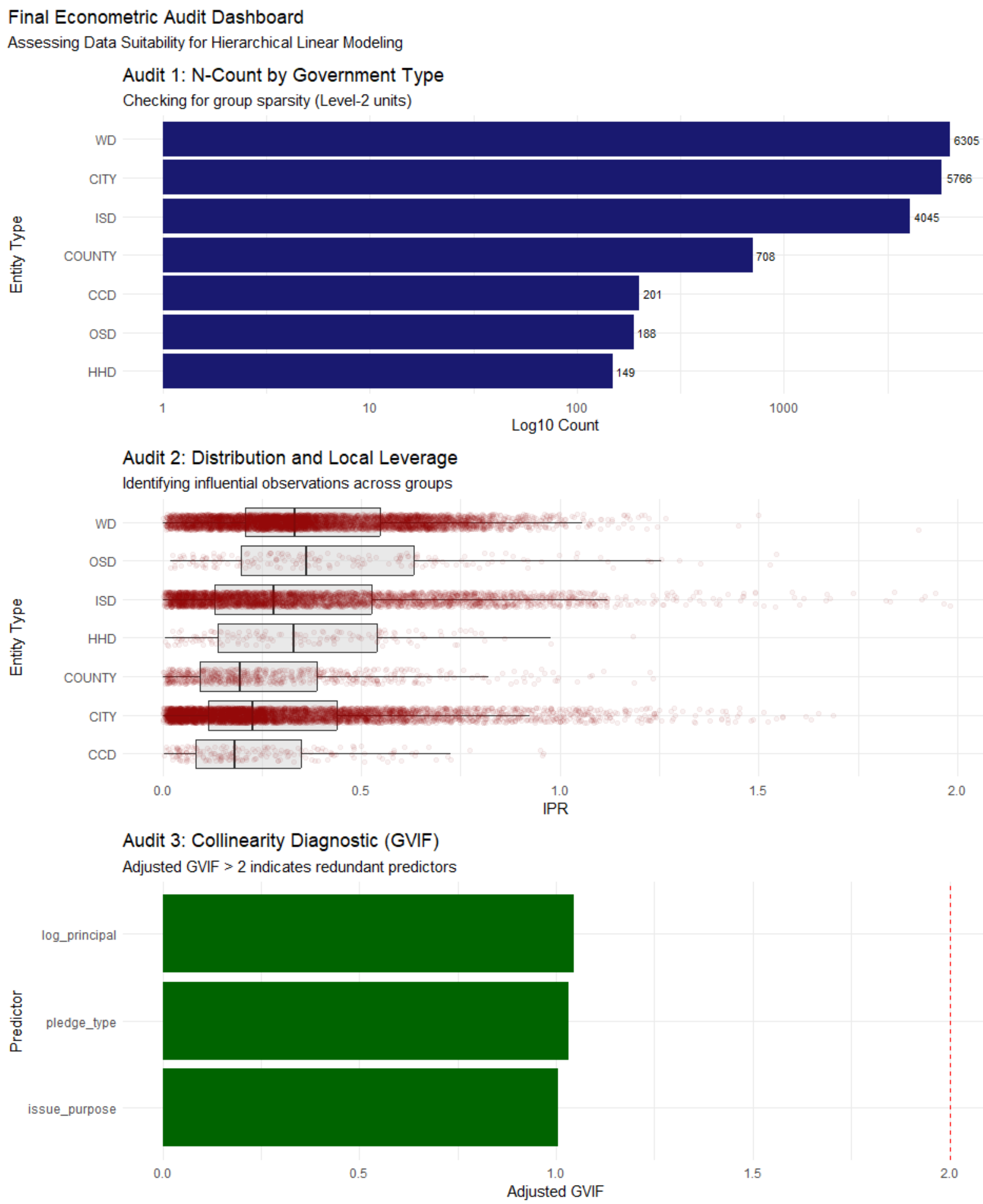


Figure 6: The Audit Dashboard

5. MODELING & EVALUATION

This section explains the statistical modeling strategy used to evaluate the structural drivers of borrowing costs in Texas municipal debt. The main objective was to compare a standard population-level regression model with a hierarchical model that accounts for institutional clustering across government types. Since the response variable, **Interest-to-Principal Ratio (IPR)**, is observed at the issuance level, but each issuance belongs to a broader government category, the data naturally has a multilevel structure.

5.1 Modeling Strategy

The first modeling decision was whether a standard OLS regression was sufficient for this dataset. OLS assumes that every observation is independent. In this project, that assumption is weak because municipal debt issuances are grouped within government types. For example, a bond issued by an Independent School District may share more similarities with other school district bonds than with a bond issued by a county or water district. These similarities may come from legal structure, tax authority, investor perception, regulatory environment, or the typical purpose of debt issued by that entity type.

Because of this clustering, OLS can provide a useful baseline, but it may underestimate uncertainty by treating every issuance as fully independent. A **Hierarchical Linear Model (HLM)** is more appropriate because it allows each government type to have its own baseline borrowing-cost level while still estimating the overall effects of pledge type, issue purpose, and principal size. This structure directly matches the project's research goal: to determine whether borrowing costs are shaped only by issuance-level variables or also by institutional identity. The project materials identify this as a key modeling issue, since government categories have different baseline levels and statistically justify a hierarchical model with group-level intercepts.

5.2 Null Hierarchical Model and ICC

The first model estimated was the **Null Hierarchical Model**, also called the unconditional model. This model does not include pledge type, issue purpose, or principal size. Its only purpose is to divide the total variation in IPR into two parts: variation between individual issuances and variation between government types.

The null model is written as:

$$\boldsymbol{IPR_{ij} = \beta_0 + u_{0j} + \epsilon_{ij}}$$

Where:

$\boldsymbol{IPR_{ij}}$ represents the interest-to-principal ratio for issuance $i$in government type j,

$\boldsymbol{\beta_0}$ is the overall mean IPR,

$\boldsymbol{u_{0j}}$ is the random intercept for government type j, and

$\boldsymbol{\epsilon_{ij}}$ is the issuance-level residual error.

The key statistic from this model is the **Intraclass Correlation Coefficient (ICC)**. The ICC measures how much of the total variance in IPR is explained by group membership. In this project, the ICC was approximately **0.062**, or **6.2%**. This means that about 6.2% of the variation in borrowing cost is attributable to **Government Type** rather than individual issuance-level characteristics. The original model output reports both adjusted and unadjusted ICC values of 0.062.

Although 6.2% may seem modest, it is meaningful in a large municipal debt dataset. It confirms that institutional identity explains a measurable portion of borrowing-cost variation. If this grouped structure were ignored, the model would treat institutional differences as random noise. Therefore, the ICC provides the statistical justification for moving from a simple regression framework to a hierarchical modeling framework.

5.3 Baseline OLS Model

After estimating the null model, a baseline **Ordinary Least Squares (OLS)** regression was fitted. The OLS model included the main fixed predictors: **Pledge Type**, **Issue Purpose**, and **Log Principal**. This model provided a population-average estimate of how these variables relate to IPR.

The OLS model can be written as:

$$\boldsymbol{IPR_i = \beta_0 + \beta_1(PledgeType_i) + \beta_2(IssuePurpose_i) + \beta_3(LogPrincipal_i) + \epsilon_i}$$

The OLS results showed that pledge type and log principal were statistically important predictors. In particular, the Revenue bond coefficient was positive and statistically significant, meaning that Revenue bonds had a higher predicted IPR than the reference category, General Obligation bonds. The OLS model also produced an adjusted $R^2$of approximately **0.3365**, meaning that the fixed predictors explained about 33.7% of the variation in IPR.

However, the main limitation of OLS is that it assumes all debt issuances are independent. It does not account for the fact that issuances from the same government type may share institutional characteristics. Therefore, OLS is useful as a benchmark, but it is not the final preferred model for this project.

5.4 Hierarchical Linear Model

The final model was a **Hierarchical Linear Model**, also known as a Linear Mixed-Effects Model. This model included the same fixed effects as OLS but added a random intercept for **Government Type**. The model was specified as:

$$IPR_{ij} = \beta_0 + \beta_1(PledgeType_{ij}) + \beta_2(IssuePurpose_{ij}) + \beta_3(LogPrincipal_{ij}) + u_{0j} + \epsilon_{ij}$$

Where:

$\boldsymbol{i}$ represents the individual debt issuance,

$\boldsymbol{j}$ represents the government type,

$\boldsymbol{u_{0j}}$ captures the government-type-specific deviation from the overall intercept, and

$\boldsymbol{\epsilon_{ij}}$ captures issuance-level unexplained variation.

The random intercept allows each government category to have its own baseline IPR. This is important because the cost structure of a county may differ from that of a city, school district, or water district. The HLM therefore models both the overall relationship between predictors and IPR and the institutional differences across government categories.

The HLM output showed that the coefficient for **Revenue bonds** was approximately **0.0587**, meaning that Revenue bonds carried about a **0.06-unit higher IPR** than General Obligation bonds after controlling for issue purpose and principal size. The model also showed that the random intercept variance for Government Type was meaningful, confirming that institutional grouping contributes to borrowing-cost differences.

5.5 Model Diagnostics

Model diagnostics were used to evaluate whether the Hierarchical Linear Model met the assumptions required for reliable interpretation. The diagnostic checks focused on residual normality, homogeneity of variance, random-effects normality, and BLUP inspection.

The residual normality plot showed that the Level-1 residuals were approximately centered around zero and followed the expected normal shape reasonably well. Although there was slight right-tail behavior, the large sample size reduces the risk that minor deviations from normality would materially affect inference.

The homogeneity of variance diagnostic showed that standardized residuals had a relatively stable spread across fitted values. There was no severe fan-shaped pattern, suggesting that the model did not suffer from strong heteroscedasticity. The log transformation of principal helped reduce variance instability common in financial debt data.

The random-effects Q-Q plot was especially important because HLM assumes that the government-type random intercepts are approximately normally distributed. The diagnostic results showed that the government-type effects fell within or near the expected confidence band, supporting the validity of the random-intercept structure.

The BLUP inspection provided an interpretable view of institutional differences. BLUPs, or Best Linear Unbiased Predictors, show how each government type deviates from the overall average after accounting for the fixed predictors. The project results show that Community College Districts and Counties had lower baseline borrowing burdens, while Independent School Districts and Water Districts had higher baseline borrowing burdens. This supports the conclusion that institutional identity is not just a background label; it directly affects borrowing-cost structure.

5.6 Final Performance Comparison

The final evaluation compared three models: the Null HLM, the baseline OLS model, and the Full HLM. The purpose was to determine whether adding fixed predictors and accounting for government-type clustering improved model performance.

Model	**Purpose**
Null HLM	Measures how much IPR variation is explained by Government Type alone
OLS	Provides a baseline population-average model
Full HLM	Final model that estimates fixed effects while accounting for Government Type clustering

The model comparison showed that Full **HLM performed better than the OLS model**. The OLS model had an AIC of approximately **-4566.1**, while the HLM had a lower AIC of approximately **-4791.9**. Since lower AIC indicates better model fit after accounting for model complexity, this result supports the HLM as the stronger model. The BIC comparison leads to the same conclusion, with HLM producing a lower BIC than OLS. HLM also achieved a slightly lower RMSE, around **0.210** compared with **0.212** for OLS.

Overall, the modeling results show that the hierarchical structure is not just a technical addition. It improves model fit, corrects for institutional clustering, and provides more meaningful interpretation of borrowing-cost differences across government categories. The final HLM confirms that Revenue bonds carry a structural premium, Government Type explains measurable variation in IPR, and municipal borrowing costs are better understood through a multilevel framework than through a simple population-average regression.

6. RESULTS AND INTERPRETATION

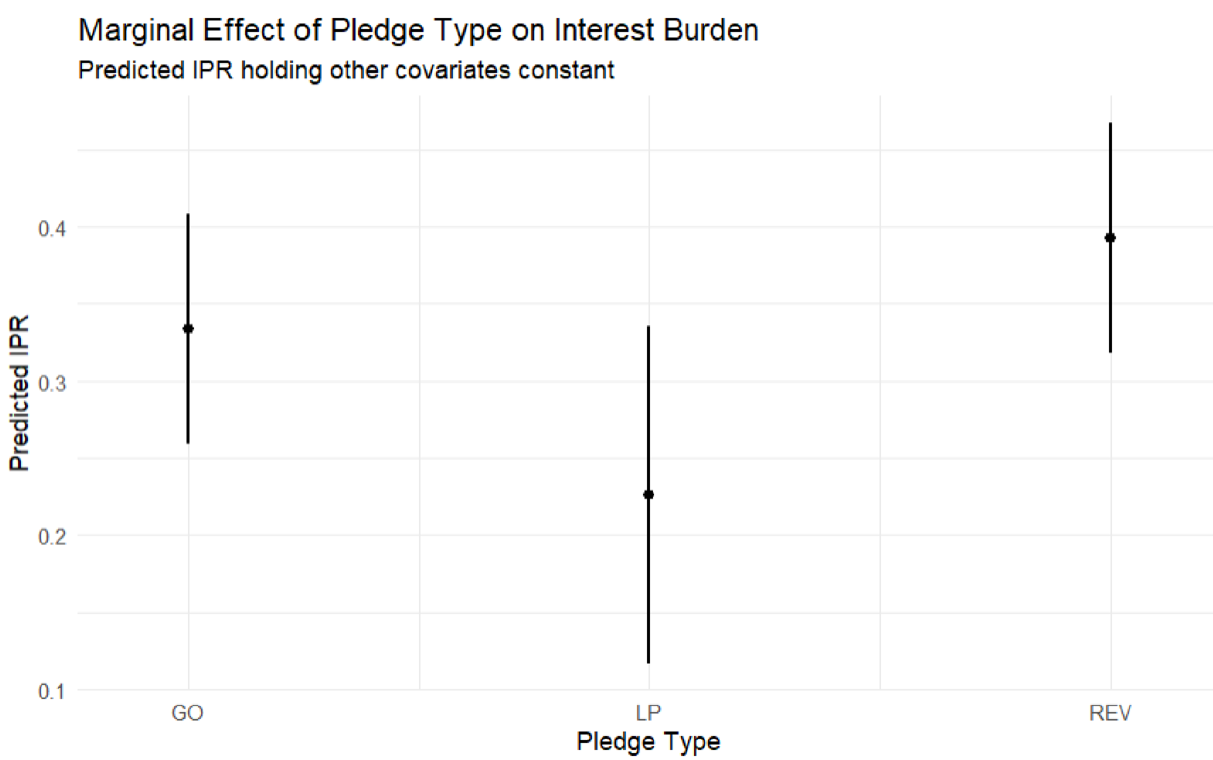
This section interprets the main statistical findings from the exploratory analysis, OLS baseline model, and final Hierarchical Linear Model. The results show that borrowing costs in Texas municipal debt are not random or purely market driven. Instead, the long-term interest burden is shaped by pledge structure, institutional identity, issue purpose, and issuance scale.

6.1 Revenue Premium

The most important finding of this project is the existence of a clear **Revenue Premium**. In the final Hierarchical Linear Model, the coefficient for **Revenue bonds** is approximately **0.0586**, meaning that Revenue bonds carry about a **0.06-unit higher Interest-to-Principal Ratio (IPR)** than General Obligation bonds after controlling for issue purpose and principal size.

This result confirms that pledge structure matters. General Obligation bonds are typically backed by the taxing authority of the issuer, while Revenue bonds depend on a narrower repayment source, such as utility revenue, toll revenue, or project-specific income. Because Revenue bonds are tied to less diversified repayment streams, investors appear to demand a higher interest burden as compensation for this additional risk.

The result is also statistically meaningful because the Revenue Premium remains visible after controlling other factors. This means the higher cost of Revenue bonds is not simply because they are larger, smaller, or issued for different purposes. The premium reflects the independent pricing effect of pledge type.



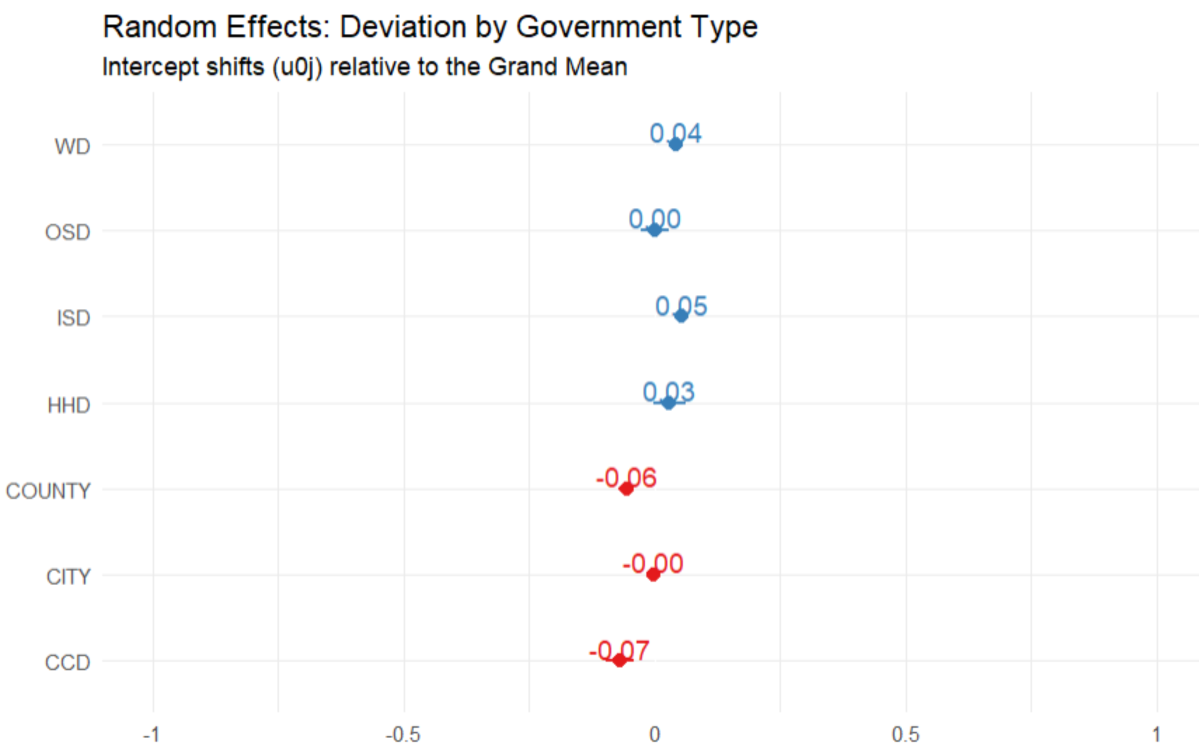
6.2 Institutional Risk by Government Type

The random effects from the Hierarchical Linear Model show that **Government Type** plays an important role in determining baseline borrowing cost. The model does not treat every issuer as identical. Instead, it allows each government category to have its own intercept, meaning each category can start from a different baseline IPR.

The results show that **Community College Districts (CCD)** and **Counties** have lower baseline borrowing burdens. In the random-effects interpretation, Community Colleges have an estimated deviation of approximately **-0.07**, while Counties have an estimated deviation of approximately **-0.06** from the grand mean. This suggests that these institutions are viewed as lower-risk borrowers, possibly because of broader statutory authority, more stable revenue structures, or stronger perceived fiscal reliability.

In contrast, **Independent School Districts (ISD)** and **Water Districts (WD)** show higher baseline borrowing burdens. ISDs have an estimated positive deviation of approximately **+0.05**, while Water Districts show approximately **+0.04**. This indicates that these entities face a structural borrowing premium even after controlling for pledge type, issue purpose, and principal size. For ISDs, this may be related to high issuance volume, while for Water Districts, it may reflect specialized infrastructure risk and narrower project-based repayment conditions.

This finding is important because it proves that borrowing cost is not only attached to the loan itself. It is also attached to the institutional identity of the borrower.



6.3 Issue Purpose Effects

Issue Purpose was included in the model to control for the reason the debt was issued. Different types of projects may carry different repayment expectations, risk profiles, and investor perceptions. The results show that some issue purposes are associated with meaningful changes in IPR.

One of the clearest findings is the negative effect of **Refunding** debt. In the final HLM, issue_purposeRefund has a coefficient of approximately **-0.169**, indicating that refunding issuances are associated with substantially lower interest burdens compared with the reference category.

This result is financially intuitive. Refunding debt is often used to refinance existing obligations under more favorable terms, reduce interest burden, or restructure debt service. Therefore, the negative coefficient suggests that Texas local governments may be using refunding issuances as an effective debt management strategy.

Some other categories, such as **Transportation** and **Toll Road**, show positive coefficients, suggesting higher borrowing burdens for certain infrastructure-heavy projects. However, the strength and precision of these effects vary by category. The broader interpretation is that issue purpose matters, but the strongest and most consistent structural result remains the Revenue Premium.

6.4 Loan Size and Market Saturation

The relationship between loan size and borrowing cost is more complex than a simple linear pattern. The EDA showed a **non-linear relationship** between **Log Principal** and IPR. At smaller and moderate issuance sizes, borrowing costs appear to decline as issuance size increases, which supports the idea of economies of scale. Larger issuances may have better liquidity, more investor attention, and lower fixed transaction costs per dollar borrowed.

However, the curve begins to rise for very large issuances. This suggests a **market saturation effect**. Extremely large debt issues may require higher interest burdens because they demand a larger pool of investors and may create concentration risk. In other words, the benefit of scale does not continue forever. At a certain point, ultra-large issuances may need to offer a higher return to attract enough capital.

The final model also shows a positive coefficient for log_principal, approximately **0.0668**, supporting the idea that larger issuances can be associated with higher IPR after other variables are controlled. The EDA therefore adds important nuance: the relationship is not simply "larger debt equals cheaper borrowing." Instead, there may be an optimal range where scale is beneficial before saturation effects begin to dominate.

6.5 Key Statistical Takeaways

The results support five major conclusions.

First, the **Revenue Premium exists**. Revenue bonds carry an estimated **0.06-unit higher IPR** than General Obligation bonds after controlling for issue purpose and principal size. This confirms that legal pledge structure has a measurable effect on borrowing cost.

Second, **institutional identity matters**. Government Type explains a meaningful share of borrowing-cost variation, and the random effects show that some entities have structural borrowing advantages while others face structural premiums. Community Colleges and Counties appear to have lower baseline costs, while ISDs and Water Districts face higher baseline burdens.

Third, **issue purpose affects borrowing burden**, especially for refunding debt. The negative refunding coefficient suggests that refinancing activity is associated with lower interest burden, which supports the role of refunding as a debt management tool.

Fourth, **loan size has a non-linear relationship with cost**. Economies of scale appear at moderate issuance sizes, but very large issuances may trigger a saturation premium. This means issuance size should be managed strategically rather than assumed to always reduce borrowing cost.

Finally, the **Hierarchical Linear Model is superior to OLS** for this dataset. The HLM accounts for government-type clustering and produces a stronger model fit, with a lower AIC than the OLS model. The OLS model has an AIC of approximately **-4566.1**, while the HLM has an AIC of approximately **-4791.9**. This confirms that Texas municipal borrowing costs are best explained using a multilevel framework rather than a simple population-average regression.

Model	AIC	BIC	RMSE
OLS	-4566.1	-4387.6	0.212
HLM	-4791.9	-4605.6	0.210

7. CONCLUSION & FINANCIAL STRATEGY

This project shows that Texas municipal borrowing costs are not driven only by broad market conditions. They are also shaped by the legal structure of the debt, the institutional identity of the issuer, the purpose of the issuance, and the size of the borrowing. By using the **Interest-to-Principal Ratio (IPR)** as the main measure of long-term debt burden, the analysis was able to compare borrowing costs across different types of local government issuances.

The most important finding is that **Revenue bonds carry a structural security premium**. Even after controlling for issue purpose and principal size, Revenue bonds show an estimated **0.06-unit higher IPR** than General Obligation bonds. This confirms that investors price Revenue bonds as riskier because repayment depends on narrower project-specific or enterprise-specific revenue streams rather than broader taxing authority. For local governments, this means the choice of pledge structure has a measurable long-term cost implication.

The analysis also confirms that **Government Type affects baseline borrowing cost**. Community College Districts and Counties show lower baseline borrowing burdens, while Independent School Districts and Water Districts show higher baseline borrowing burdens. This suggests that the market does not view all public issuers equally. Institutional identity acts as a risk signal, influencing borrowing costs even after accounting for observable loan characteristics.

Another key finding is that issuance size does not have a simple linear relationship with cost. The EDA showed that moderate increases in debt size may create economies of scale, but very large issuances can face a **market saturation effect**. Extremely large debt offerings may require higher interest burdens to attract enough investor demand, meaning that "larger" is not always cheaper.

From a financial strategy perspective, local governments should evaluate debt structure more carefully before issuance. If a project can reasonably support a General Obligation structure, it may reduce long-term borrowing burden compared with a Revenue-backed structure. Issuers should also consider the scale of the borrowing and avoid assuming that larger issuances always produce lower costs. In some cases, splitting debt into more strategically timed issuances may reduce saturation pressure. Finally, refunding opportunities should be monitored closely because the model shows that refunding debt is associated with lower interest burden, making refinancing timing an important tool for debt-cost management.

Overall, the Hierarchical Linear Model provided a stronger and more realistic framework than standard OLS because it captured both loan-level effects and institutional differences across government types. The final conclusion is that Texas municipal borrowing costs are structurally patterned, not random. Effective debt planning should therefore account for pledge type, institutional category, issuance size, and refinancing timing rather than relying only on general market averages.

8. FUTURE WORK

Future work can strengthen this project by adding more financial, macroeconomic, geographic, and methodological depth to the current hierarchical modeling framework.

First, future versions should include **bond credit ratings** from agencies such as **S&P, Moody's, and Fitch**. The current model captures pledge type and government type, but it does not directly include issuer credit quality. Adding ratings would help separate whether the Revenue Premium is caused by the legal pledge structure itself or by weaker credit profiles among Revenue-backed issuers.

Second, the model should include macroeconomic variables such as the **Federal Funds Rate**, **inflation/CPI**, and broader interest-rate indicators. The EDA showed a sharp increase in borrowing costs after 2021, so adding macroeconomic controls would allow the model to distinguish structural borrowing-cost differences from broader market-rate shocks.

Third, future work should explore **geographic clustering**. Texas has large regional differences between urban markets, suburban districts, rural counties, and specialized utility districts. Adding regional indicators or nested geographic effects could show whether borrowing costs differ across areas such as DFW, Houston, Austin, San Antonio, and rural regions.

Fourth, the model can be improved by testing **nonlinear and interaction effects**. The current analysis shows that issuance size may have a non-linear relationship with IPR, so a quadratic term for log principal or spline-based approach may capture the market saturation effect more accurately. Interaction terms such as pledge_type × government_type or pledge_type × fiscal_year could also test whether Revenue bonds are penalized more heavily for certain government categories or during high-rate periods.

Fifth, future work should compare **fixed effects and random effects** approaches more formally. The current HLM uses Government Type as a random intercept, but a fixed-effects model could be tested as an alternative to evaluate whether the conclusions remain stable when government categories are treated as non-random group indicators.

Finally, future versions should explore alternative response-variable specifications, such as a **log-transformed IPR model** or a **Gamma Generalized Linear Mixed Model (Gamma GLMM)**. Since IPR is positive, continuous, and right-skewed, these models may better fit the distributional structure of the response variable and improve inference for high-interest-burden issuances.

Overall, future work should move the project from a strong structural borrowing-cost model toward a richer municipal credit-risk framework that combines legal structure, institutional identity, macroeconomic timing, geography, and credit quality.

9. REFERENCES

1) E. F. Fama and K. R. French, "Common risk factors in the returns on stocks and bonds," *Journal of Financial Economics*, vol. 33, no. 1, pp. 3–56, 1993.
2) J. Capeci, "Local fiscal policies, default risk, and municipal borrowing costs," *Journal of Public Economics*, vol. 53, no. 1, pp. 73–89, 1994.
3) B. R. Moulton, "Random group effects and the precision of regression estimates," *Journal of Econometrics*, vol. 32, no. 3, pp. 385–397, 1986.
4) J. J. Hox, *Multilevel Analysis: Techniques and Applications*, 2nd ed. New York, NY, USA: Routledge, 2010.
5) K. P. Burnham and D. R. Anderson, *Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach*, 2nd ed. New York, NY, USA: Springer, 2002.
6) Texas Bond Review Board, *Debt Outstanding by Issuance — Local Dataset*, Texas Bond Review Board, 2026.
7) Municipal Securities Rulemaking Board, "Sources of repayment," MSRB.
8) D. Bates, M. Mächler, B. Bolker, and S. Walker, "Fitting linear mixed-effects models using lme4," *Journal of Statistical Software*, vol. 67, no. 1, pp. 1–48, 2015.